

Exploratory Data Analysis (EDA) Report

Objective: To analyze the sales dataset and uncover meaningful patterns, trends, and business insights using statistical analysis, SQL queries, and data visualization techniques.

Dataset Overview

- Total Rows: 1000
- Total Columns: 12
- Dataset Source: ApexPlanet Sales Dataset

Descriptive Statistics

Summary statistics were calculated for Quantity, Unit Price, and Total Sales.

Key Statistics:

- Average Quantity Sold: 5.44
- Average Unit Price: ₹25,486.78
- Average Total Sales: ₹139,399.44
- Maximum Quantity Sold: 10
- Maximum Unit Price: ₹49,997.53
- Maximum Total Sales: ₹493,677.50
- Minimum Quantity Sold: 1
- Minimum Unit Price: ₹145.78
- Minimum Total Sales: ₹437.34

Univariate Analysis

Visualizations Created:

- Histogram of Total Sales
- Histogram of Quantity
- Category Distribution Bar Chart

Findings

- Electronics generated the highest revenue among all categories.

- Sales values are widely distributed, ranging from ₹437 to ₹493,677.
- Most transactions involve quantities between 3 and 8 units.

SQL Business Analysis

Business questions answered:

1. Top 5 Products by Revenue
2. Revenue by Category
3. Average Sales by Category
4. Top Customers by Revenue
5. Monthly Revenue Trend
6. Quantity Sold by Category
7. Top Customers by Orders

Revenue by Category

	Category	Revenue
0	Electronics	50778581.70
1	Education	25031689.40
2	Grocery	22231711.28
3	Furniture	21521561.48
4	Fashion	19835895.79

Top Products by Revenue

	Product	Revenue
0	Laptop	25443008.51
1	Mobile	25335573.19
2	Book	25031689.40
3	Rice	22231711.28
4	Chair	21521561.48

Monthly Revenue Trend

Top Revenue Months:

1. March – ₹13,059,899.94
2. June – ₹12,912,332.64
3. November – ₹12,627,620.22

Lowest Revenue Month:

- September – ₹9,179,896.29

Correlation Analysis

Techniques Used:

- Scatter Plot
- Correlation Heatmap
- Pair Plot

Correlation Matrix Findings

- Quantity vs Total Sales = 0.6466
- Unit Price vs Total Sales = 0.6863
- Quantity vs Unit Price = 0.0219

Interpretation

- Total Sales has a strong positive relationship with both Quantity and Unit Price.
- Quantity and Unit Price are almost independent of each other.
- Unit Price shows the strongest influence on Total Sales.

Dashboard KPIs

- Total Revenue: ₹139,399,439.65
- Total Orders: 1000
- Average Order Value: ₹139,399.44
- Total Categories: 5
- Best Performing Category: Electronics

Key Insights

1. Electronics is the highest revenue-generating category.
2. Laptop is the top-selling product by revenue.
3. March recorded the highest monthly sales revenue.
4. Higher quantities and higher unit prices directly increase total sales.
5. Unit Price has the strongest impact on revenue generation.

Conclusion

The sales dataset was successfully analyzed using descriptive statistics, SQL querying, and exploratory data analysis techniques. Electronics and Laptop products contributed the highest revenue, while March emerged as the best-performing month. The analysis revealed strong positive relationships between Total Sales, Quantity, and Unit Price, providing valuable business insights for future decision-making.